

NPTEL Course on "Computer System Optimization for ML"

Lecture #40

NPTEL

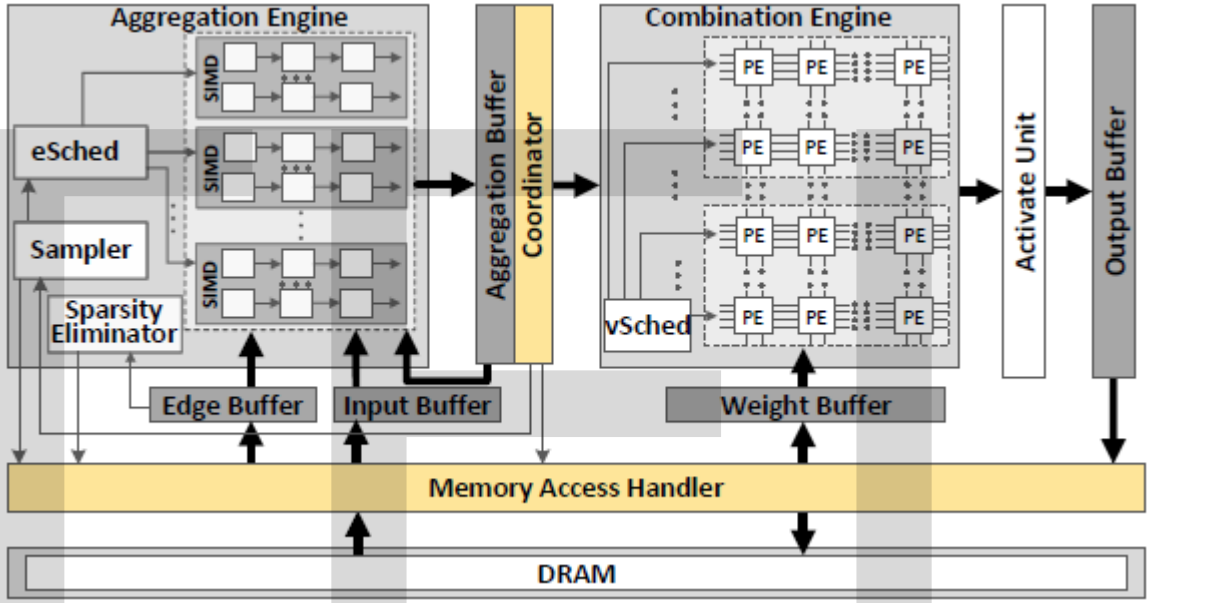
Agenda

- HyGCN
- Basics of Analog In-memory Computing
- ISAAC

NPTEL

HyGCN Architecture [1]

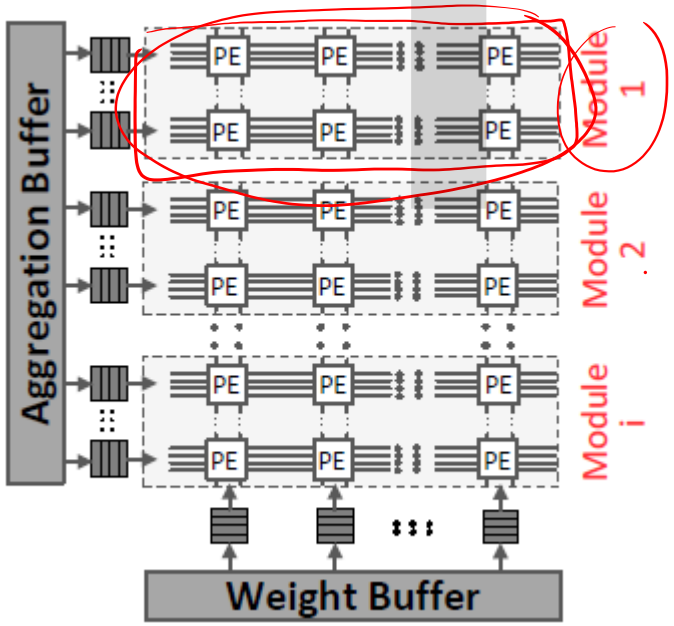
- HyGCN
  - A hybrid architecture
  - Two engines: Aggregation and Combination
  - Memory access Handler
  - A communication interface (coordinator)



[1] Yan, Mingyu, et al. "Hygcn: A gcn accelerator with hybrid architecture." 2020 IEEE International Symposium on High Performance Computer Architecture (HPCA). IEEE, 2020.

Combination Engine

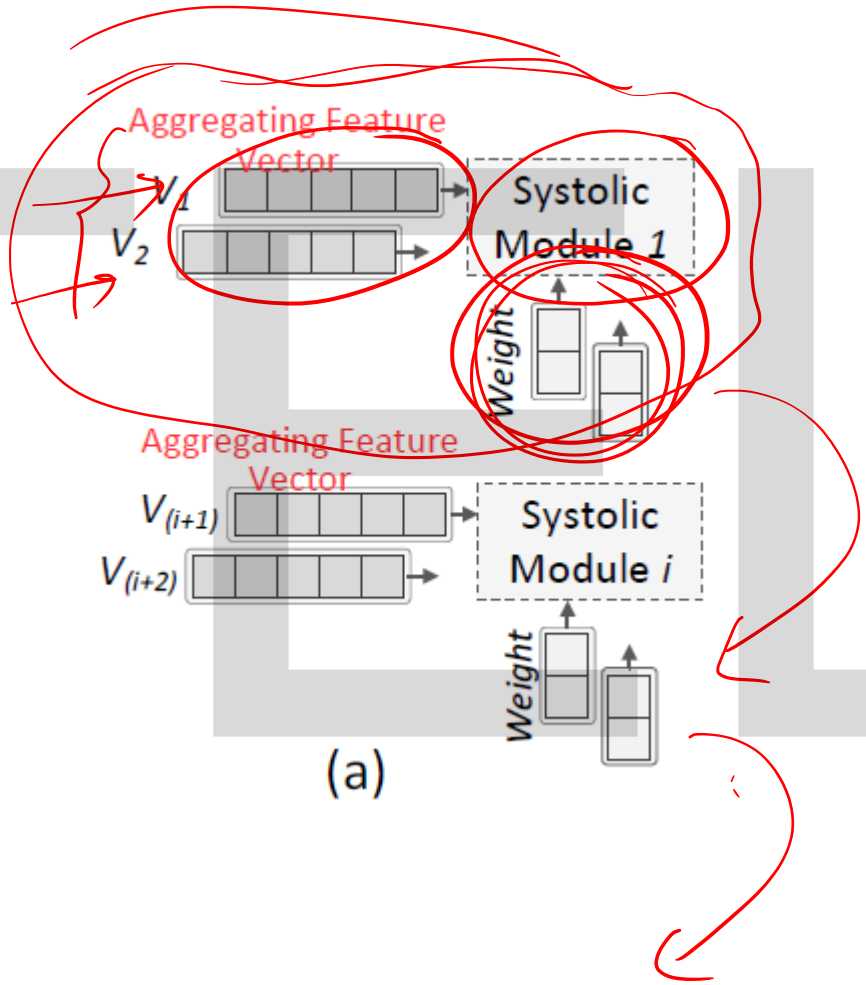
- Combination operation at each vertex acts like a neural network
- The execution is regular but compute-intensive
- Systolic array-based design
- Multiple arrays are integrated rather than a single one
- A group of systolic arrays is assembled to form a systolic module



NPTEL

Working Modes of the Combination Engine

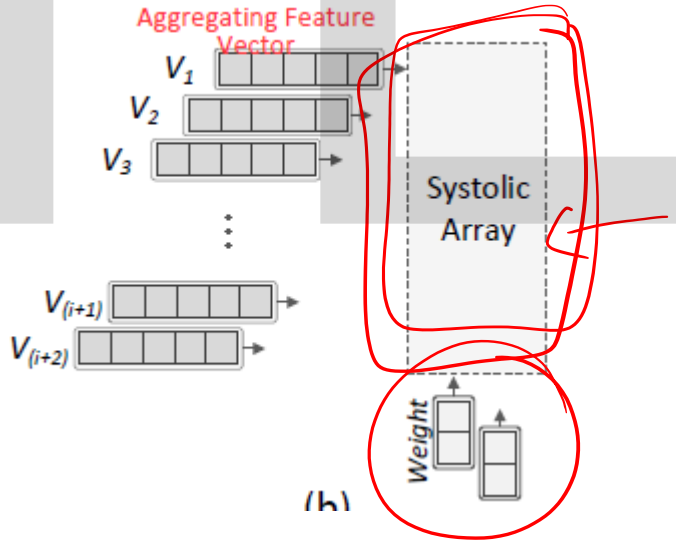
- **Independent working mode**
  - Systolic modules work independently from each other
  - Each of them processes the MVM operations of a small group of vertices
  - The weight parameters are directly accessed from the Weight Buffer
  - Lower vertex latency since combination can be done as soon as the aggregated features are ready
  - Goes well with the vertex dispersed mode
    - Aggregated features are produced quickly and sequentially



NDTL

Working Modes of the Combination Engine

- **Co-operative Working Mode**
  - **Systolic modules are further assembled together to simultaneously process more vertices**
  - **The mode requires to assemble the aggregated features of a large group of vertices together before performing the combination operation**
  - **More reuse of weight**



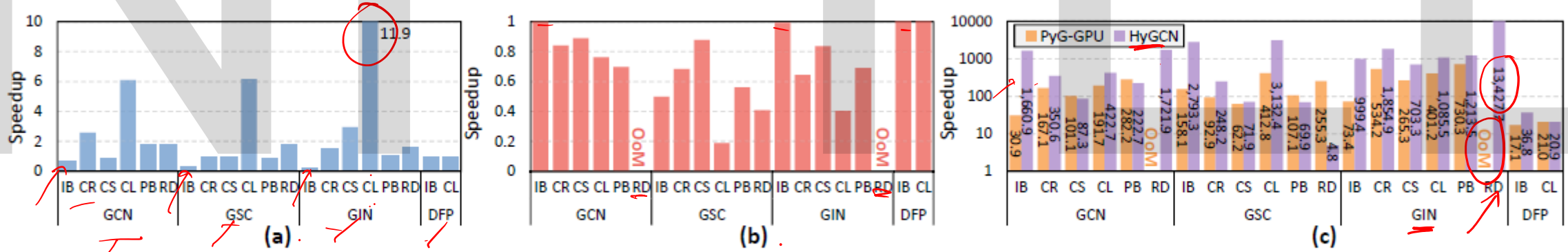
NPTEL

Evaluation Results: Speed Up

SYSTEM CONFIGURATIONS.

	PyG-CPU	PyG-GPU	*HyGCN*
Compute Unit	2.5 GHz @ 24 cores	1.25Ghz @ 5120 cores	1 GHz @ 32 SIMD16 cores and 8 systolic modules (each with 4×128 arrays)
On-chip Memory	60MB	34MB	128 KB (Input), 2 MB (Edge), 2 MB (Weight), 4 MB (Output) and 16 MB (Aggregation)
Off-chip Memory	136.5GB/s DDR4	~900GB/s HBM~2.0	256GB/s HBM~1.0

Note: GPU's on-chip memory includes the register files, and L1 and L2 caches.



Comparistion to PyG-CPU and PyG-GPU: Speedup of our algorithm optimization on (a) CPU and (b) GPU; (c) Speedup over the optimized PyG-CPU. OoM means the evaluation fails in running on GPU due to out of memory.

Evaluation Results: Data Access

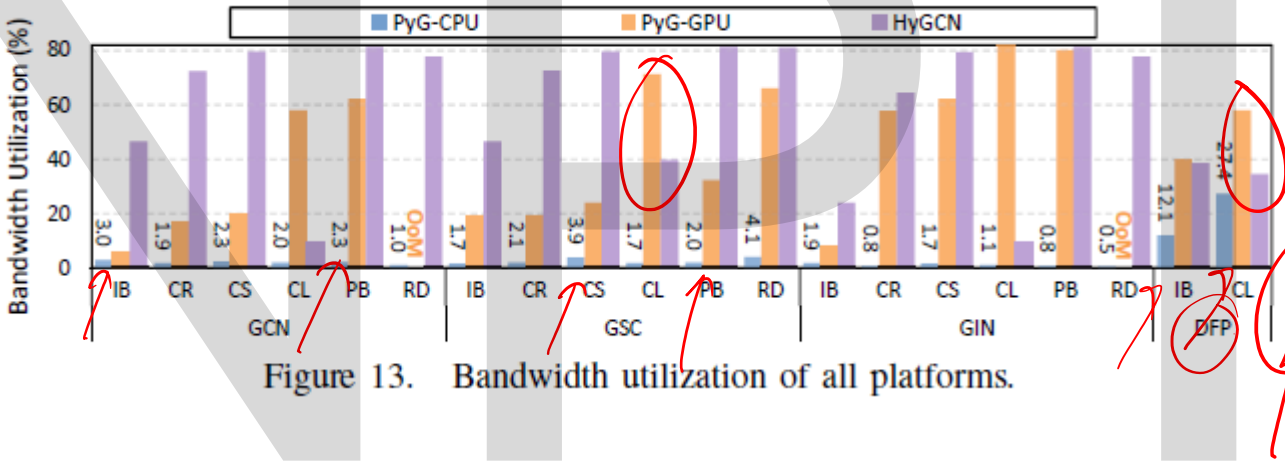


Figure 13. Bandwidth utilization of all platforms.

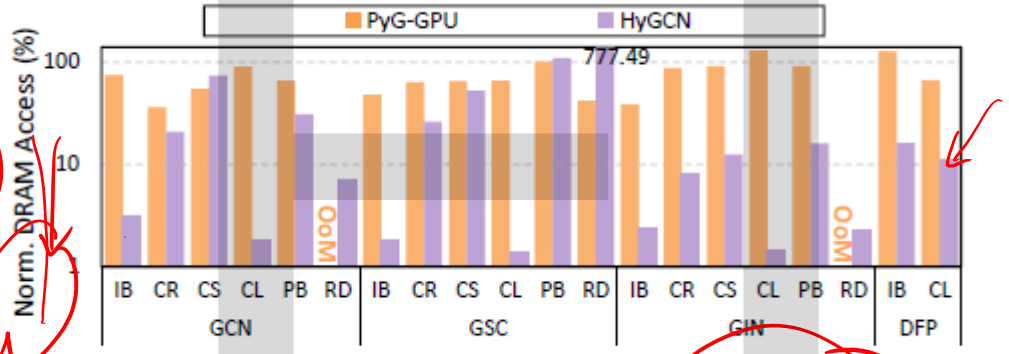


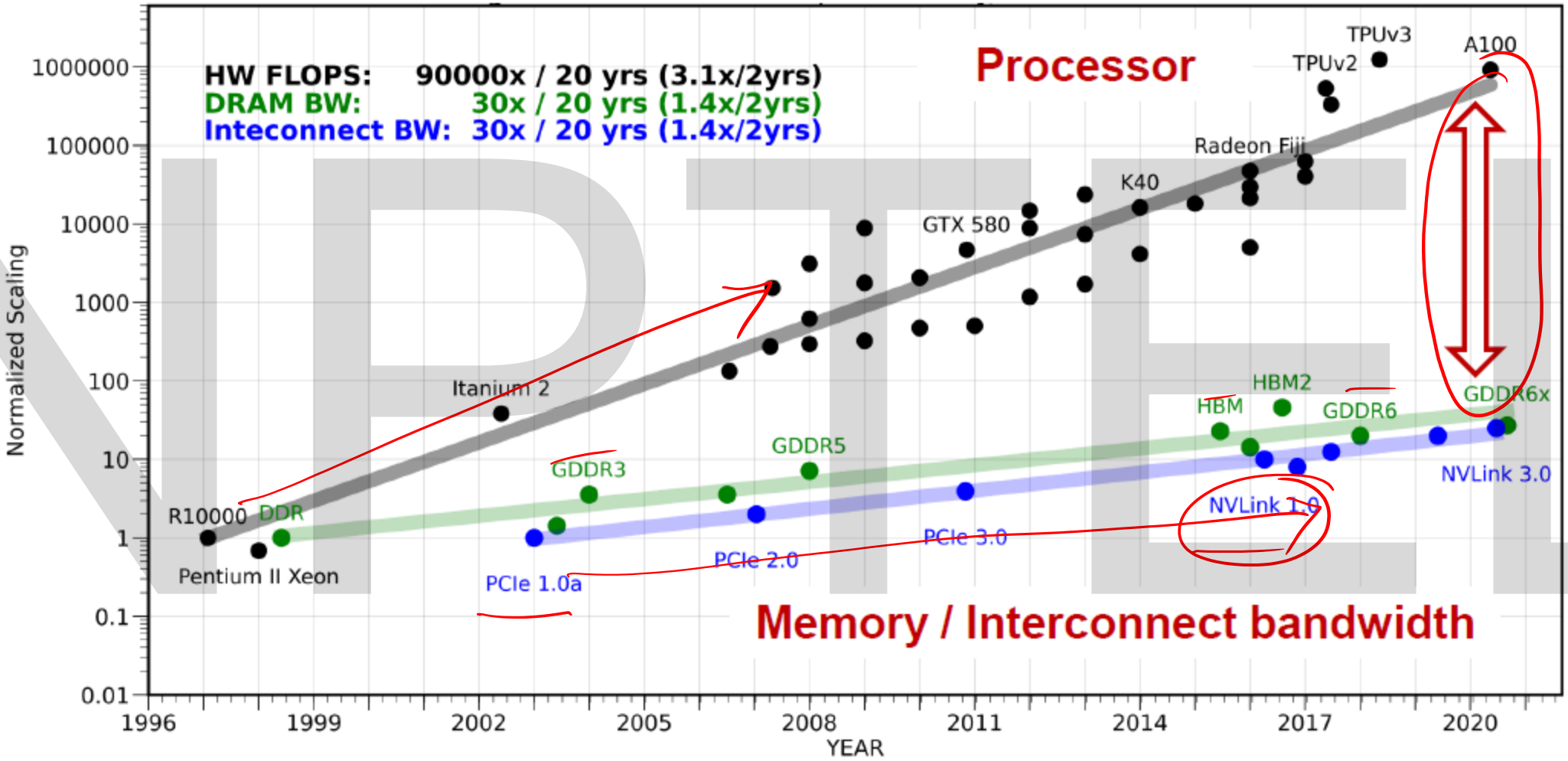
Figure 14. Normalized data access to PyG-CPU.

Agenda

- HyGCN
- **Basics of Analog In-memory Computing**
- ISAAC

NPTEL

Need for In-Memory Computation



[1] Work by A. Gholami, 2020; Hynix; Chipworks; Created by Prof. Yu Cao, ASU.

NDTL

Agenda

- HyGCN
- Basics of Analog In-memory Computing
- **ISAAC**

NPTEL



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)